Step 1: Empirical Card Sorting Experiment

1.1 Card Deck Preparation

For this experiment, I focused on the Weber State eWeber class registration system that I audited in Phase 1. One of the main problems I found was that several actions were confusing or difficult to find, especially when students were trying to add or drop classes.

I asked different users to organize the registration features into the order that they found to be best.

I created 18 cards showing the most important actions students might need while registering for classes.

Card #	Content Item / Task
1	Search for Classes
2	Browse Available Courses
3	Filter Classes by Subject
4	View Class Details
5	Check Class Availability
6	View Prerequisites
7	Add a Class
8	Drop a Class
9	Review Pending Classes
10	Submit Registration
11	View Registration Confirmation
12	View Current Schedule
13	Check Schedule Conflicts
14	View Waitlist Status
15	Find Course Reference Number (CRN)
16	View Registration Dates
17	Check Registration Holds
18	View Tuition and Fees

I chose these items because they cover the main registration process along with related information that students might need before or after registering.

1.2 Participant Execution

For the experiment, I used the open card sorting method meaning participants could create their own categories and name them instead of choosing from some options that I provided.

The three participants were James, a college student familiar with university websites but not Weber State registration as he goes to the U, Jake, my dad who is familiar with registration, and Amber,my mom with about the same experience using the registration site as my dad, but not familiar with registration terms.

Each participant was given the same 18 cards and asked to organize them into groups that made sense to them and name each group. The results are below.

James's Card Sorting Results

Category Label	Cards
Find Classes	1, 2, 3, 4, 5, 6, 15
Register for Classes	7, 9, 10, 11
My Schedule	8, 12, 13
Registration Information	14, 16, 17, 18

James grouped most of the course information under Find Classes. He felt that students should be able to see prerequisites, availability, and course numbers while searching instead of having to visit another page. He separated adding from dropping because dropping usually starts from an already made schedule.

Jake's Card Sorting Results

Category Label	Cards
Course Search	1, 2, 3, 4, 5, 6, 15
Registration Actions	7, 8, 9, 10, 11
Schedule	12, 13
Registration Status & Dates	14, 16, 17
Financial Information	18

Jake grouped Add and Drop together because he viewed them as parts of the same registration process. Unlike James, he placed tuition and fees separately because he felt financial information shouldn’t be with class registration info.

Amber's Card Sorting Results

Category Label	Cards
Look for Classes	1, 2, 3, 4, 5, 6, 15
Sign Up for Classes	7, 9, 10, 11
My Classes	8, 12, 13, 14
Important Information	16, 17, 18

Amber used more casual category names, such as “Sign Up for Classes” instead of “Registration Actions.” She placed waitlist status under My Classes because she viewed being on a waitlist as part of managing her schedule.

1.3 Data Synthesis and Analysis

After comparing the three participants, I saw that there is a pretty good pattern of where the main registration features belonged. All three grouped Search for Classes, Browse Available Courses, Filter Classes, View Class Details, Check Availability, View Prerequisites, and Find CRN together.

There were more disagreements about dropping classes, checking waitlist status, and viewing tuition information.

Participant Summary

Content Group	Participant Agreement	Finding
Course Search (1–6, 15)	3 of 3	Strong agreement that course information belongs together
Add / Review / Submit / Confirm (7, 9–11)	3 of 3	Students use these when completing registration
Current Schedule / Conflicts (12–13)	3 of 3	Both should be found together
Drop Class with Schedule (8, 12)	2 of 3	Most expect dropping a class from their schedule page
Waitlist with Registration Dates (14, 16)	2 of 3	Some saw waitlists as registration status
Dates / Holds / Tuition (16–18)	2 of 3	Most grouped these as information to check before registering

Similarity Matrix

This table shows how often two cards were placed in the same category out of the 3 participants.

Card Pair	Agreement
Search Classes + Browse Courses	3/3 (100%)
Search Classes + View Prerequisites	3/3 (100%)
Search Classes + Find CRN	3/3 (100%)
Add Class + Submit Registration	3/3 (100%)
Review Pending + Confirmation	3/3 (100%)

Current Schedule + Schedule Conflicts	3/3 (100%)
Drop Class + Current Schedule	2/3 (67%)
Waitlist Status + Registration Dates	2/3 (67%)
Registration Dates + Registration Holds	3/3 (100%)
Registration Holds + Tuition and Fees	2/3 (67%)

1.4 Findings and Mental Models

The biggest takeaway was that all three participants wanted course information grouped together. This supports my earlier idea that students should not need to switch between different pages just to check whether a class is available or if they meet its prerequisites.

One thing that surprised me was the disagreement about dropping classes. I thought that they would be together because the original system combines them. However, James and Amber both placed Drop Class with their schedule instead which surprised me.

This makes sense when I think about it from the user's perspective. Adding a class starts with finding something new, while dropping a class starts with something already on their schedule. But I think personally I would've chosen differently, showing that these results could change with a larger sample size.

Another difference was how participants named their categories. Jake used more technical names, while Amber used more casual names. This difference in naming should help me get an idea of how to name options in a way that both explains the action but is also casual enough for new users.

Step 2: Revised Sitemap & Information Architecture Blueprint

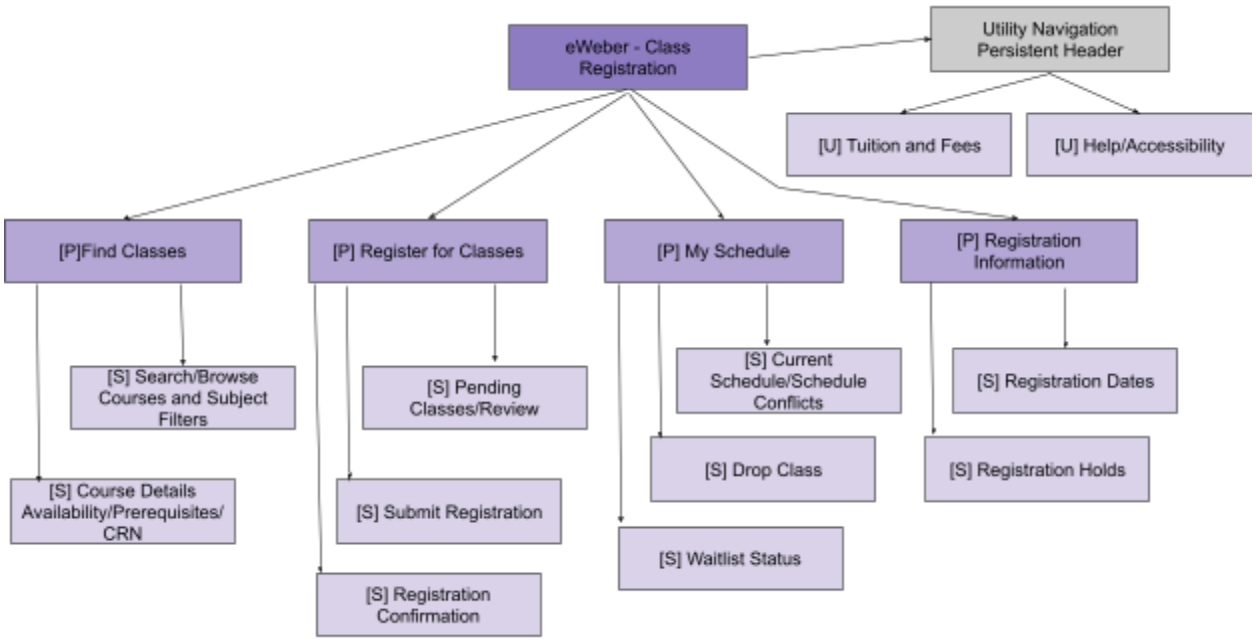
2.1 IA Restructuring

For this step, I used the card sorting results to decide how the class registration website should be organized. All three grouped the class search information together, and they also kept adding, reviewing, and submitting classes in the same general area. I wanted the new navigation to follow those patterns instead of making students figure out where the website puts everything.

The biggest change is separating Find Classes, Register for Classes, and My Schedule. Two participants placed Drop Class with their current schedule rather than with adding a class, so I put dropping a class under My Schedule. I also kept registration dates and holds together under Registration Information. Tuition and Fees will stay available as a utility link, since it is useful but not part of the main registration task.

2.2 Revised Sitemap

The sitemap below shows the navigation and the pages or actions available under each section. I kept the structure fairly shallow so most information is one selection away from the main page.



Legend: [P] Primary navigation, [S] Secondary page or task, [U] Utility navigation.

2.3 Navigation Hierarchy

Navigation Level	Items	Reason
Primary	Find Classes, Register for Classes, My Schedule, Registration Information	Matches the main task groups from the card sort.
Secondary	Course Details, Pending Classes, Confirmation, Current Schedule, Drop Class, Waitlist, Dates, Holds	Keeps related pages inside the section students would expect.
Utility	Tuition and Fees, Help / Accessibility	Makes supporting information available

		without mixing it into the main registration flow.

2.4 Global and Contextual Navigation

Type	Placement / Elements	Expected Behavior
Global navigation	Persistent header with Find Classes, Register for Classes, My Schedule, and Registration Information	Students can switch between main areas without returning to the homepage.
Utility navigation	Header utility links for Tuition and Fees and Help / Accessibility	Supporting links remain available from every page.
Contextual navigation	Breadcrumbs such as Find Classes > Course Details	Shows where the student is and provides a clear way back.
Inline actions	Add Class on course details, Review Pending Classes after adding, Drop Class beside a registered course	Places the next action near the information it is related to.
Status feedback	Pending count in registration area, Registered confirmation after submission	Makes the difference between selecting a class and registering for it clear.

2.5 Reducing Navigation Depth

I wanted to avoid making users open many menus just to find important information. Important details should be available directly from Course Details while searching. Similarly, students should be able to open My Schedule and drop a registered class from there instead of searching for a separate Add/Drop utility.

This structure still gives the registration process its own review and confirmation steps, just as pages instead of having everything shoved into one interface. This should hopefully make things feel simpler.

Step 3: User Flows

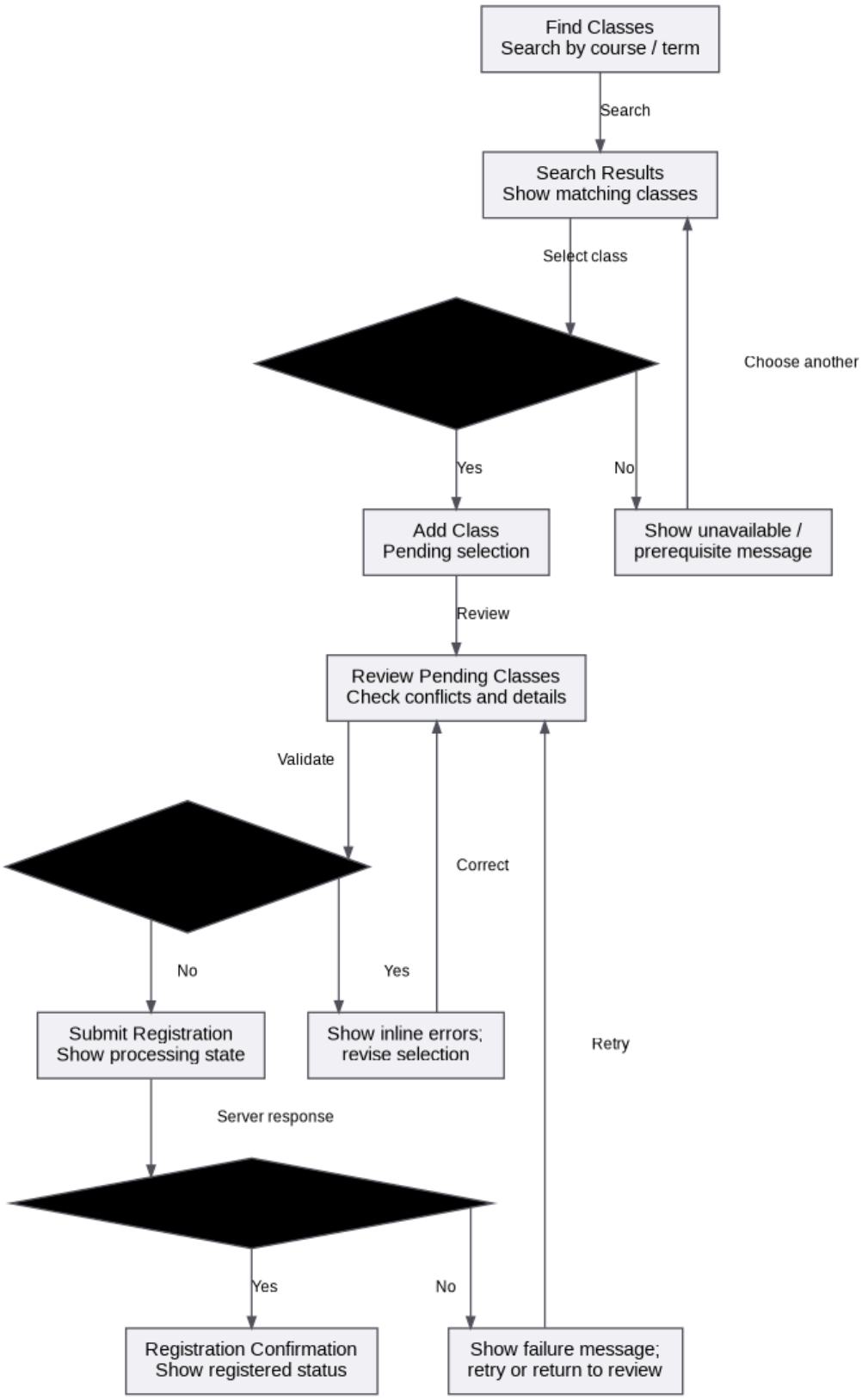
Weber State eWeber — Revised Class Registration System

3.1 Flow Selection

I chose two tasks from the Phase 1 audit: registering for a class and dropping an existing class. These were important because the original registration process made it difficult to tell whether a class was only added as a pending selection or actually registered. Dropping a class also needs a clear confirmation so students do not accidentally remove something from their schedule.

3.2 User Flow 1: Register for a Class

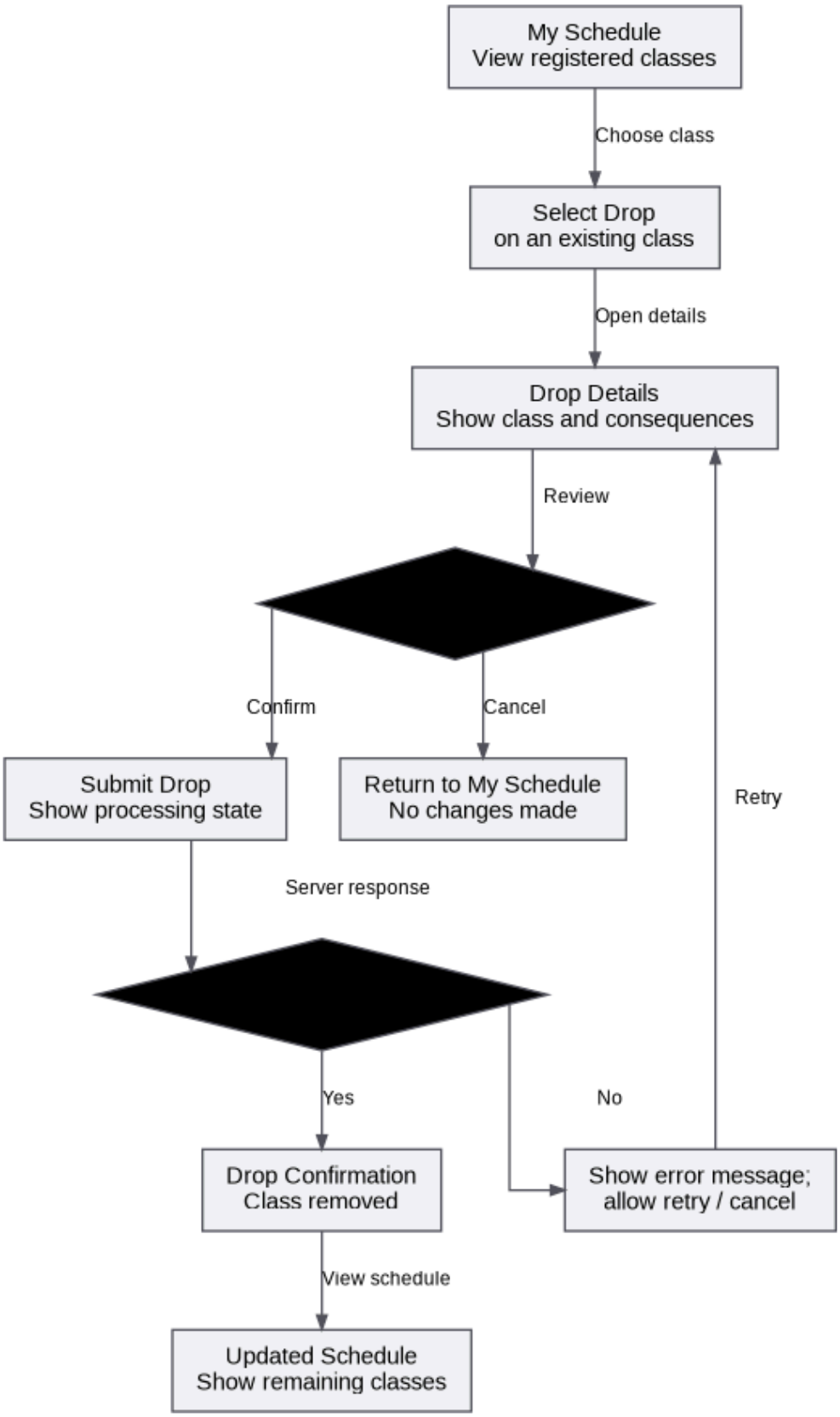
This flow begins in Find Classes and ends with a registration confirmation. Students can check availability and prerequisites while viewing results. Classes are marked as pending until registration is submitted and confirmed by the system.


Find Classes
Search by course / term
Search
Search Results
Show matching classes
Select class
Choose another
Yes
No
Add Class
Pending selection
Show unavailable /
prerequisite message
Review
Review Pending Classes
Check conflicts and details
Validate
Correct
No
Yes
Submit Registration
Show processing state
Show inline errors;
revise selection
Retry
Server response
Yes
No
Registration Confirmation
Show registered status
Show failure message;
retry or return to review

Decision points: If a class is unavailable or prerequisites are not met, the student returns to the results. If the review finds a conflict, the student corrects the selection before submitting. If submission fails, the system displays an error and keeps the student's selections for another attempt.

3.3 User Flow 2: Drop an Existing Class

This flow starts in My Schedule rather than Find Classes. The student selects an existing class, reviews the drop details, and confirms the action. The schedule only changes after the system confirms the drop was successful.


My Schedule
View registered classes
Choose class
Select Drop
on an existing class
Open details
Drop Details
Show class and consequences
Review
Confirm
Cancel
Submit Drop
Show processing state
Return to My Schedule
No changes made
Retry
Server response
Yes
No
Drop Confirmation
Class removed
Show error message;
allow retry / cancel
View schedule
Updated Schedule
Show remaining classes

Decision points: The student can cancel without changing their schedule. If the drop request fails, the system displays an error and allows another attempt. A successful request leads to a confirmation and then an updated schedule.

3.4 Screen Transitions and System Responses

Flow	Main screen transitions	Important system feedback
Register	Find Classes -> Results -> Review Pending -> Confirmation	Availability/prerequisite messages, conflict validation, processing, success or failure
Drop	My Schedule -> Drop Details -> Drop Confirmation -> Updated Schedule	Cancellation, processing, success or failure, updated class status

Both flows keep the outcome of each action clear so that the user never has to wonder if a class was added, registered, or dropped.

3.5 Wireframe Deck Creation

Link to Figma Wireframe:
RegistrationWireframe

Step 4: Accessibility First & Semantic Navigation Planning

4.1 Semantic HTML Structuring

I would use the same basic page structure on each screen so students do not have to relearn the navigation when moving between Find Classes, Review, and My Schedule. The main content changes, but the landmarks stay consistent.

HTML element	Placement in the wireframe	Purpose

<header>	Top bar: Weber State / Class Registration title	Identifies the page and provides a consistent starting point
<nav aria-label="Primary">	Main navigation: Find Classes, Register for Classes, My Schedule, Registration Information	Lets users move between the four main areas
<nav aria-label="Breadcrumb" >	Breadcrumb row directly above the page heading	Shows where the current screen fits in the flow
<main id="main-content">	Central content area, one per page	Contains the task the student is completing
<section aria-labelledby="…">	Search form, results, pending classes, confirmation, or schedule details	Groups related content under visible headings
<aside aria-labelledby="…">	Optional right-side summary of pending classes or registration status	Provides supporting information without interrupting the main task
<footer>	Bottom help and accessibility links	Keeps supporting links in a consistent location

4.2 Keyboard Tab Order Blueprint

The table below shows the intended Tab sequence for the controls. Static headings, results text, and other noninteractive content are read by assistive technology but aren't going to be separate Tab stops.

Flow 1 — Register for a Class

Screen	**Keyboard Tab sequence**	**Keyboard / system behavior**
1. Find Classes	1 Skip link -> 2 Primary navigation links -> 3 Term -> 4 Search input -> 5 Subject filter -> 6 Search button -> 7 Footer help links	Enter activates Search, visible focus follows the on-screen order.
2. Search Results	1 Skip link -> 2 Primary navigation -> 3 Breadcrumb link -> 4 Edit search -> 5 Filter controls -> 6 Class details buttons -> 7 Add	Class details can expand with Enter/Space, after Add, a status message announces the update.

	buttons -> 8 Review Pending Classes -> 9 Footer links	
3. Review Pending Classes	1 Skip link -> 2 Primary navigation -> 3 Breadcrumb link -> 4 Remove buttons -> 5 Back to Results -> 6 Submit Registration -> 7 Footer links	If the list is empty, Submit is disabled, an error summary explains the issue.
4. Registration Confirmation	1 Skip link -> 2 Primary navigation -> 3 View My Schedule -> 4 Find More Classes -> 5 Footer links	A confirmation heading receives focus, the confirmation number is shown as text.

Flow 2 — Drop an Existing Class

Screen	Keyboard Tab sequence	Keyboard / system behavior
1. My Schedule	1 Skip link -> 2 Primary navigation -> 3 Term selector -> 4 Drop buttons for each class -> 5 Footer links	Each Drop button has an name that includes its class.
2. Drop Details	1 Skip link -> 2 Primary navigation -> 3 Breadcrumb link -> 4 Cancel -> 5 Continue to Confirmation -> 6 Footer links	Class name and potential consequences appear before the confirmation action.
3. Confirm Drop	1 Skip link -> 2 Primary navigation -> 3 Breadcrumb link -> 4 Cancel -> 5 Confirm Drop -> 6 Footer links	Confirmation is explicit, while processing, duplicate submission is prevented.
4. Updated Schedule	1 Skip link -> 2 Primary navigation -> 3 View Remaining Classes -> 4 Find Classes -> 5 Footer links	A success message is announced, and the updated schedule remains available.

4.3 ARIA Landmarks and State Attributes

Component	HTML / ARIA plan	When it applies

Primary navigation	<nav aria-label="Primary">, active link aria-current="page"	All screens
Breadcrumbs	<nav aria-label="Breadcrumb" >, current item aria-current="page"	Results, review, drop details, confirmation
Expandable class details	<button aria-expanded="false" aria-controls="details-10 1">	Results screen, changes to true when opened
Search and filters	Visible <label for="…"> associated with each input	Find Classes and Results
Validation errors	aria-invalid="true" and aria-describedby="…" on invalid fields	Missing required search input or registration issue
Status updates	for added class, successful update	Nonurgent messages without moving keyboard focus
Submission errors	or focused error summary	Registration/drop failure requiring attention
Disabled submit	Native disabled attribute on button when action is unavailable	Empty pending list or while processing
Supporting summary	<aside aria-labelledby="summar y-heading">	Review and schedule summary when present

Step 5: Post-Mortem Reflection

5.1 How did the empirical card sorting results challenge your initial assumptions about the navigation structure?

One thing that surprised me during the card sorting was how differently participants viewed certain registration tasks. I originally assumed that adding and dropping classes should be grouped together because that is how the current eWeber system handles

them. However, James and Amber both placed dropping a class under their existing schedule instead of registration.

This made me realize that just because two actions are technically related doesn't mean users think of them the same way. Adding a class starts with searching for something new, while dropping a class starts with managing something already on your schedule.

The results did show that keeping course information together was the preferred format. All three participants grouped prerequisites, availability, and course details with searching for classes. All of these results could change with a larger sample size, but overall, I can see how this would help a lot with designing a layout that users expect.

5.2 How do ARIA landmarks and keyboard tab order improve navigation for screen-reader users?

ARIA landmarks and a logical keyboard tab order help make a website easier to navigate without relying on a mouse or visual layout.

For example, using HTML elements like <header>, <nav>, and <main> allows screen-reader users to jump directly to important sections instead of having to go through everything on the page. Attributes like aria-current="page" also help users understand where they are in the page.

Keyboard tab order is important because users should be able to move through buttons, links, and form fields in a predictable sequence. In my design, I planned the tab order to follow the same general order as the visual layout, so users can move through searching, selecting, and registering without getting lost.

I think these changes are important because accessibility should be part of the original design rather than something added afterward. They also make the interface more organized and easier to understand for everyone. Both users and future developers who are working on improving the pages.